Laboratory report of Nanjing University of Information Science and Technology

Experiment Title Cloud-Native Automated Deployment Data
Class Applied Computing Cooperative 1
Name Lu Moyou ID 202483910013

一、Experimental purpose

The primary goal of this experiment is to introduce students to the core concepts of Cloud Computing through hands-on practice with Platform-as-a-Service (PaaS) and Automation.

二、Experiment Contents

This lab consists of five integrated modules designed to simulate a professional cloud deployment lifecycle.

三、Detailed steps of the experiment

Lab Report Requirements: Clearly present the procedure step by step, with corresponding screenshots provided for each stage. Screenshots must demonstrate that they were taken on your own computer, for example by including visible system time, desktop environment, or other identifiable personal elements.

Step 1: Resource Acquisition

Log in to GitHub.

Navigate to the repository:

https://github.com/yhldrf/cloud-computing-lab-github

and click Fork.

Cloud Concept: You are creating a private instance of the project in the cloud ecosystem.

Step 2: Enabling the Hosting Service

Go to Settings > Pages.

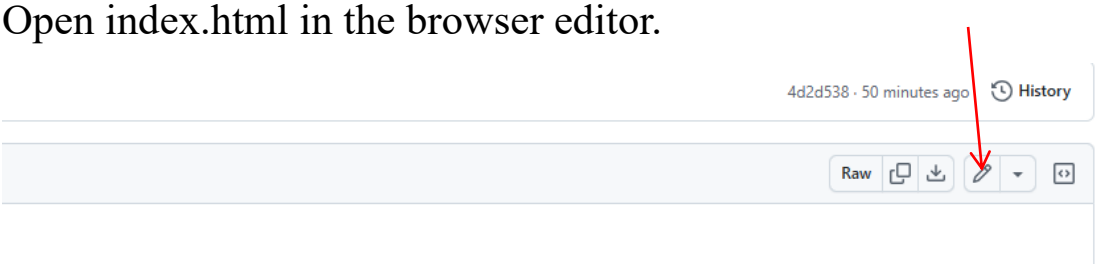
Under Build and deployment > Source, select "GitHub Actions".

After that, leave this page.

Cloud Concept: You are switching from "Manual Upload" to "Managed Pipeline Service."

Step 3: Triggering Cloud Automation

Open index.html in the browser editor.

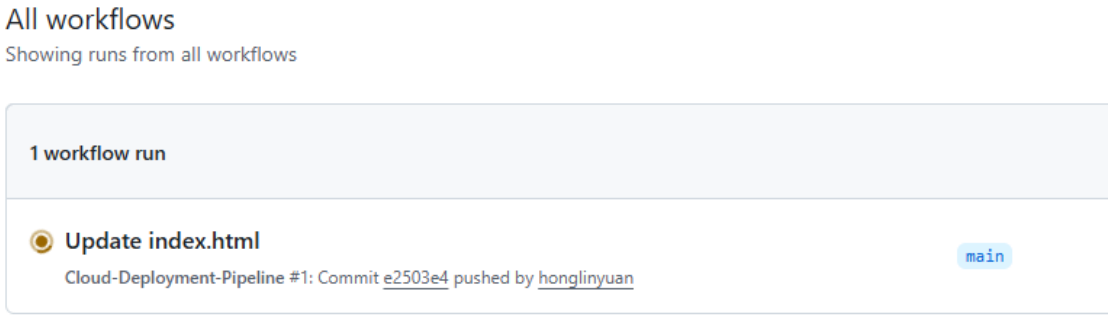


Change [Enter Your Name Here] to your actual name.

Click Commit Changes.

Immediately click the Actions tab at the top of the page.

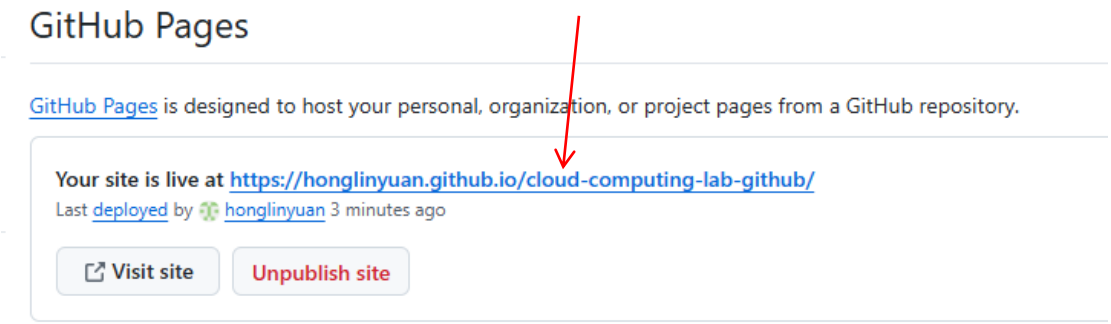
Observation: Watch the "Cloud-Deployment-Pipeline" start.



Cloud Concept: GitHub is spinning up a Virtual Machine (Ubuntu) to process your request. This is "On-demand Computing."

Step 4: Verification & Global Access

Once the Action finishes (green check), go back to Settings > Pages to find your Live URL.



Open the URL on your laptop.

Cloud Concept: The site is deployed via a PaaS (Platform as a Service) model. The cloud provider manages the underlying infrastructure and global delivery, allowing us to focus only on our content.

Step5: Content Personalization & Elements (optional)

Task: Beyond changing your name, modify the following HTML elements to see how the cloud updates different types of content.

✓ Change the Theme Color (CSS Styles)

Find the<style>block at the top and change the background color or font size.

Action:

```
<style>
    body {
        font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
        text-align: center;
        padding-top: 50px;
        background-color: #f4f4f9; /* Add a light grey background */
    }
    .status { color: #007bff; font-weight: bold; } /* Change status to Blue */
</style>
```

Cloud Concept: You are modifying the User Interface (UI) settings stored in the cloud.

✓ Multimedia Integration (Images)

Demonstrate how the cloud handles external resources.

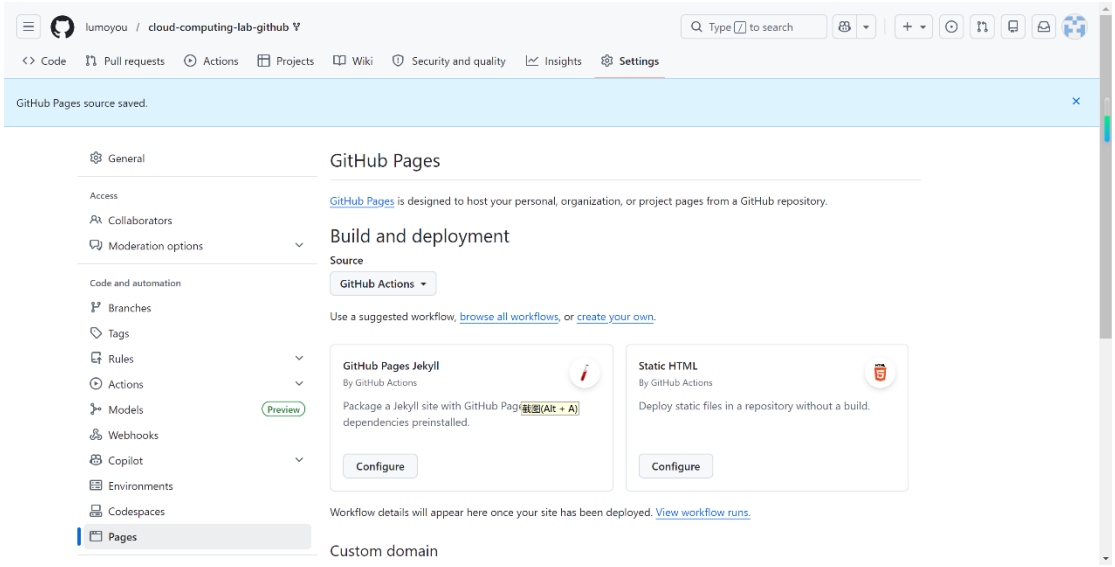
What to add: An<img>tag inside the<body>.

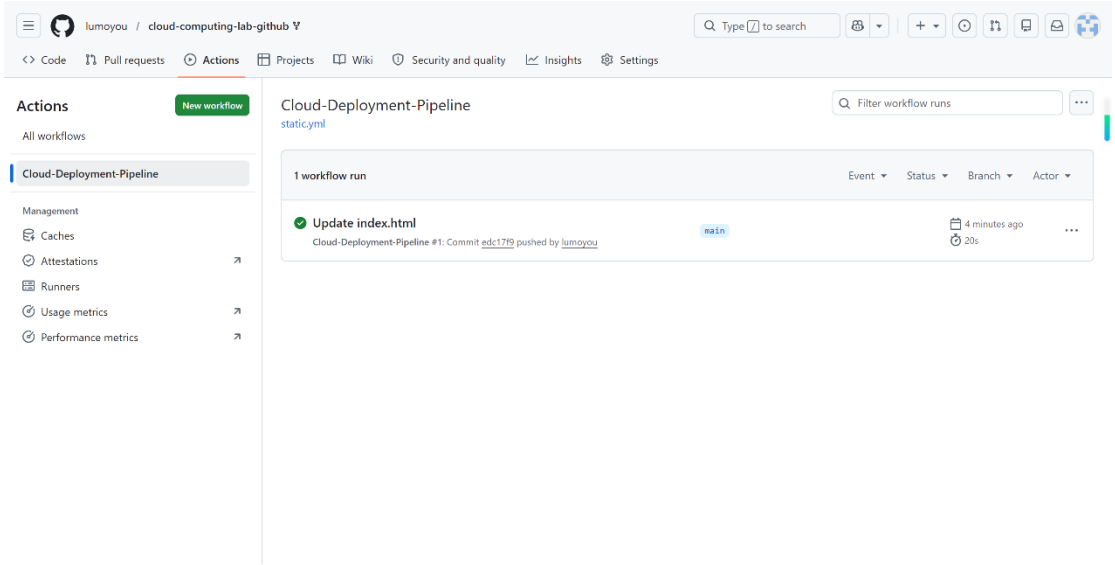
Example:

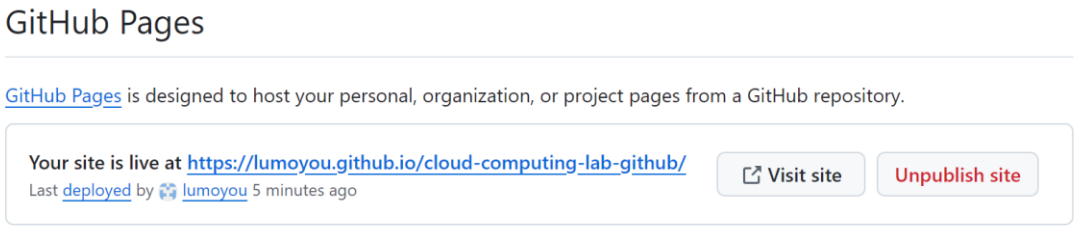
```
<img
src="https://github.githubassets.com/images/modules/site/features/actions-icon-actions.svg" width="100" alt="Actions Logo">
```

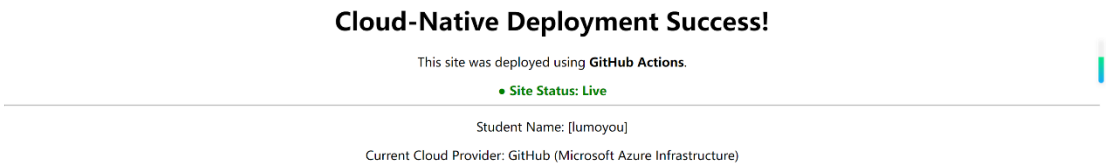
Observation: Notice that as soon as you click "Commit," the Actions tab triggers a new build automatically. This is Continuous Deployment (CD).

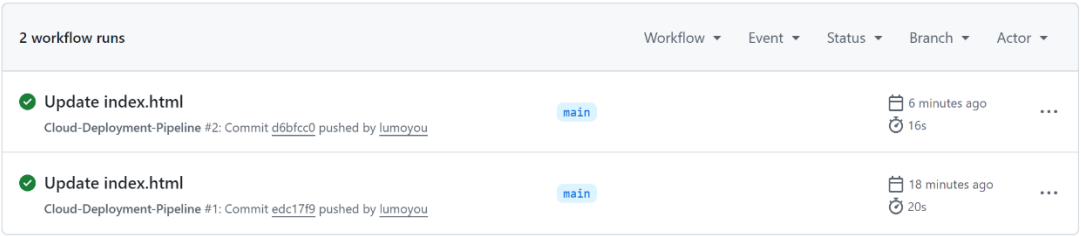
Screenshots:

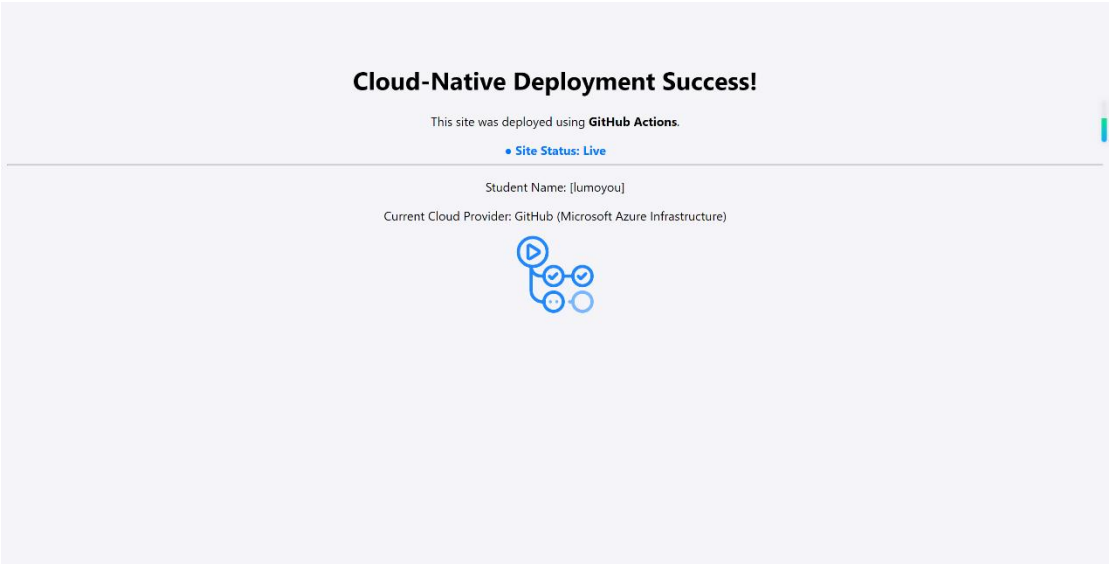












四、Reflection

1. Where is your website stored?

My website is stored on GitHub's cloud servers using GitHub Pages and GitHub Actions.

2. Did you have to install any software?

No, I didn’t install any software. I only used a browser and GitHub’s online editor.

3. Problems Encountered and Proposed Solutions

- Problem: Deployment failed.

Solution: Check HTML code for errors and re-commit.

- Problem: URL not updated.

Solution: Wait for Actions to finish and clear browser cache.